CARS DATASET ANALYSIS

(MINOR PROJECT REPORT)

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1. Dataset overview

The dataset contains information about people, their cars, countries, and credit card types. It contains 30,000 rows and 8 columns.

Column	Description	Data type
First_Name	Person's first name	Categorical text
Last_Name	Person's last name	Categorical text
Country	Country associated with the record	Categorical text
Car_Brand	Brand of the car	Categorical text
Car_Model	Model of the car	Categorical text
Car_Color	Colour of the car	Categorical text
Year_of_Manufacture	Year the car was manufactured	Integer
Credit Card Type	Type of credit card	Categorical text

Objective: To understand the dataset's structure, examine data quality, study the distribution of car brands and manufacturing years, compare brands across countries, and identify useful patterns through visualizations.

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2. Data-quality checks

The notebook includes checks for missing values, duplicate records, unique values, and outliers.

Check	Result or finding
Total records	30,000
Total columns	8
Missing values	0 in every column in the original dataset
Exact duplicate rows	Checked in the notebook; see the executed output for the count

Unique car brands	78
Unique car models	1,006
Unique countries	230
Unique car colours	19
Unique manufacturing years	65
Data types	7 text columns and 1 integer column

The notebook's written analysis also reports extra whitespace in four country values. These values should be standardized before comparing country names.

Observation: The dataset appears complete because the original columns contain no missing values. However, completeness does not automatically guarantee that every category or value is correct.

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3. Before and after data cleaning

The notebook uses mode imputation, median imputation, an IQR-based outlier treatment, and label encoding.

Data-quality aspect	Before cleaning	After cleaning
Missing values	0 reported	0 reported
Duplicate records	Checked; original count should be verified from the output	No duplicate-removal step is shown
Country whitespace	Extra spaces reported in four country values	Explicit whitespace removal is not shown
Manufacturing-year outliers	Checked using a box plot and IQR	Values clipped to the calculated lower and upper bounds
Categorical columns	Stored as text	Converted to integer labels using LabelEncoder
Dataset size	30,000 rows, 8 columns	The notebook reports the same 30,000 rows and 8 columns

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4. Descriptive statistics

Descriptive statistics help summarize the dataset and understand its main characteristics.

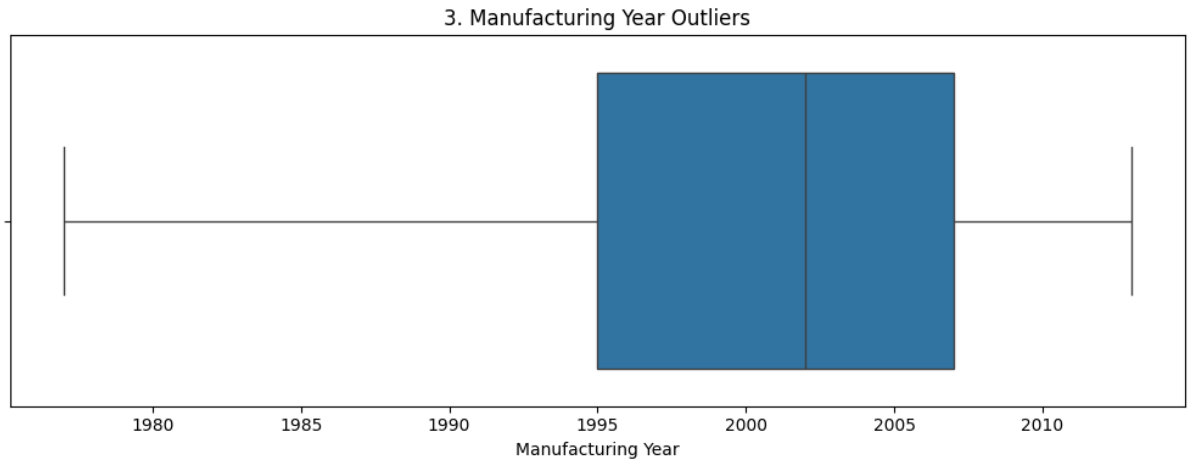
The notebook's written analysis reports the following findings:

Statistic	Result
Total car records	30,000
Distinct car brands	78
Distinct car models	1,006
Distinct countries	230
Distinct car colours	19
Mean manufacturing year	Approximately 2000
Median manufacturing year	2002
First quartile (Q1)	1995
Third quartile (Q3)	2007

The middle 50% of manufacturing years fall between 1995 and 2007. The mean is approximately 2000, while the median is 2002.

CHART:

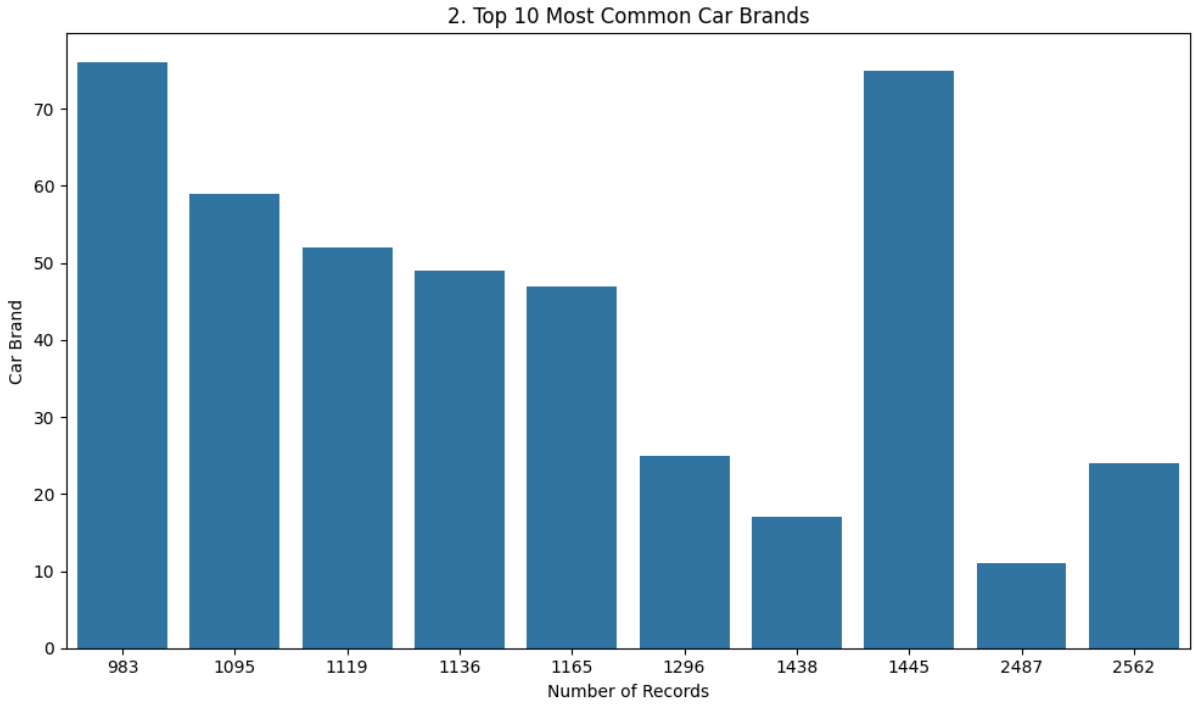
BoxPlot:



Purpose: Shows the median, quartiles, spread, and potential outliers.

Observation: The IQR method is used to identify unusually low or high manufacturing years. The before-and-after plots should be compared to evaluate the effect of clipping.

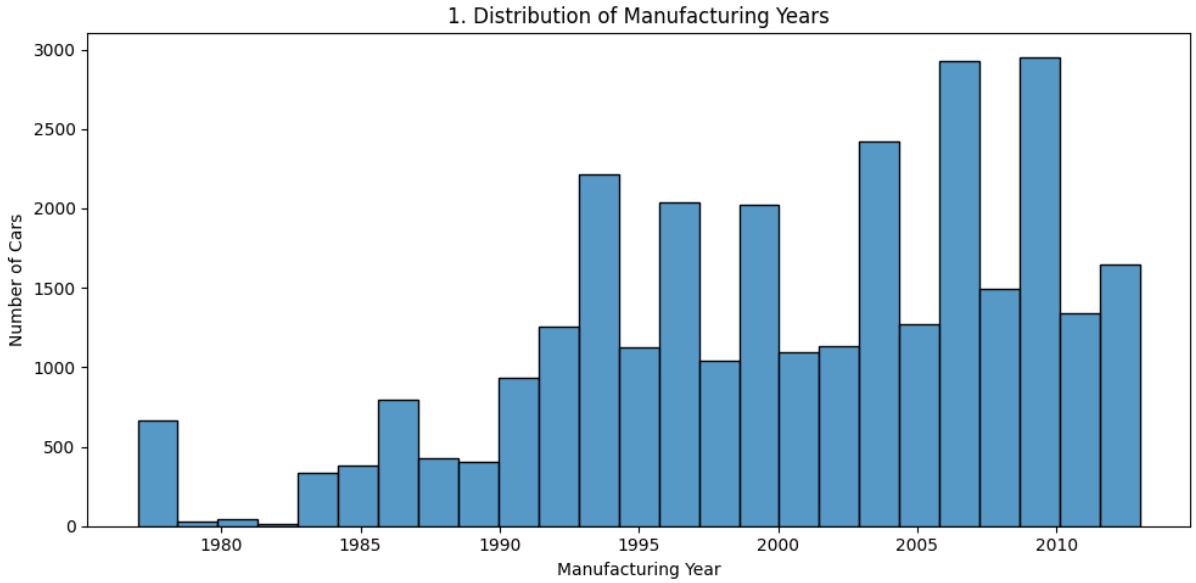
BarChart:



Purpose: Compares the number of records for the most common brands.

Observation: Ford is the most frequently recorded brand, with 2,562 records, followed by Chevrolet with 2,487 records.

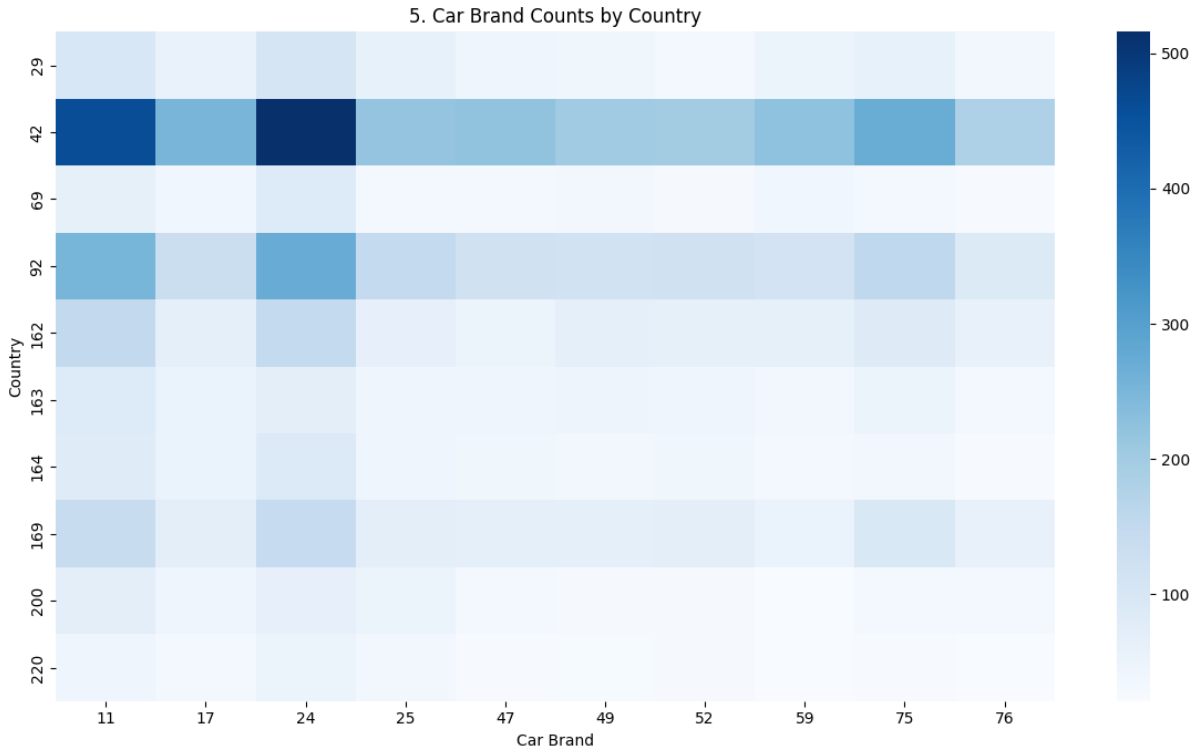
Histogram:



Purpose: Shows how car records are distributed across manufacturing years.

Observation: The reported mean is around 2000 and the median is 2002. The histogram can show whether the records cluster around particular years.

HeatMap:



Purpose: Compares the frequency of selected car brands across the ten most represented countries.

Observation: The notebook reports China as the country with the most records. The heatmap helps compare brand frequencies within the selected countries.

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7. Key insights

- Ford and Chevrolet lead the recorded brands. Together, they account for 5,049 records, or approximately 16.83% of the dataset.
- The records cover many countries. The 230 distinct country values indicate broad geographical variety, although the distribution is uneven.
- No single car model dominates. The most frequent model represents only about 0.70% of all records.
- Manufacturing years are concentrated around the early 2000s. The mean and median suggest that this period is central to the dataset.
- Categorical data needs suitable analysis. Brand and country comparisons are best presented using counts and categorical charts, rather than treating label-encoded values as continuous numerical measurements.

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8. Limitations

1. The dataset contains no information about car prices, mileage, engine capacity, or vehicle condition, limiting the analysis of vehicle performance and value.
2. The records may not represent the actual distribution of cars worldwide. Country counts describe this dataset only.
3. The dataset's source and collection methodology are not established by the notebook, so its representativeness cannot be confirmed.
4. The presence of names and credit card types does not establish a relationship between payment methods and car ownership.
5. Label encoding introduces arbitrary integer codes. Numerical plots or correlations using these codes may be misleading.
6. Clipping manufacturing years can conceal unusual but valid records. The original values and calculated IQR limits should be documented.
7. The notebook's country whitespace finding should be addressed explicitly before final country-level analysis.

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9. Conclusion

The exploratory data analysis examined 30,000 records across eight columns, focusing on car brands, models, colours, countries, and manufacturing years. The notebook identified Ford as the most frequent brand, China as the country with the most records, and a manufacturing-year distribution centred around the early 2000s.

Data-quality checks found no missing values in the original dataset. The analysis also included IQR-based outlier treatment and label encoding of categorical variables. These steps demonstrate common data-preprocessing techniques, although category standardization and the effects of cleaning should be verified explicitly.

Overall, the project demonstrates how exploratory data analysis can help summarize a dataset, identify patterns, inspect potential data-quality issues, and communicate findings through charts. The results should be interpreted as patterns within the available records rather than definitive conclusions about the global car market.

REFERENCE:

CARS DATASET FROM KAGGLE.

LINK: https://www.kaggle.com/code/ahmedessamsaber/cars-dataset/input